Milestone 3: Report

Coronavirus Tweet Categorization and Exploration

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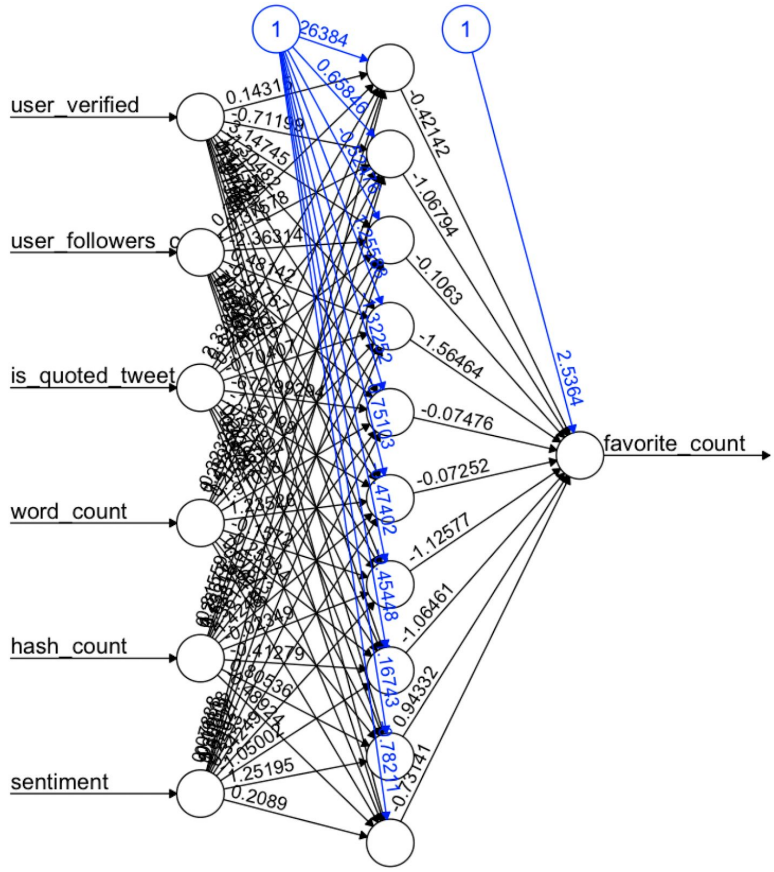
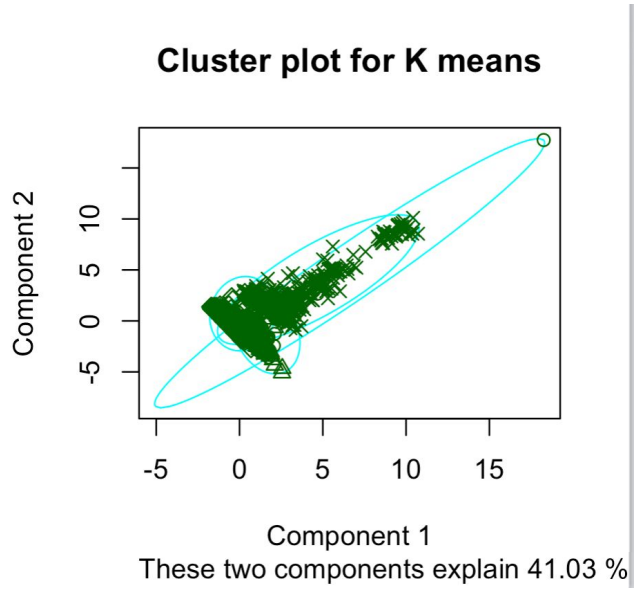
RESEARCH QUESTION 1 - Predicting Tweet Favorites and Retweets

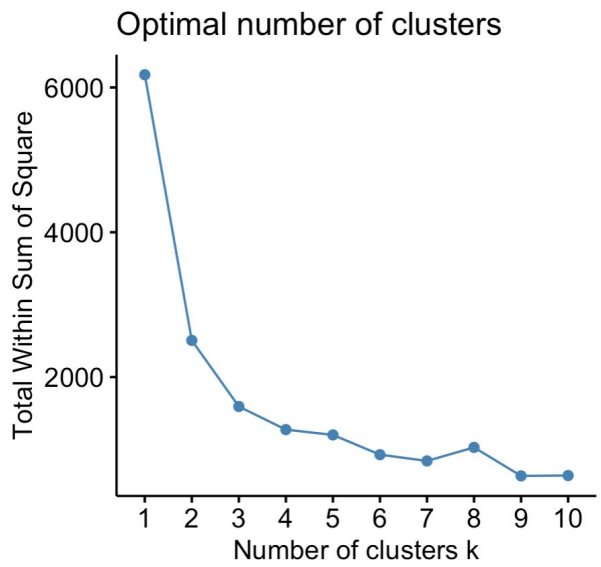
- **Tweet Favorites**
  - **Q1: SVR**
    - Part A
      - Since this is a regression problem a SVR model had to be created instead of an SVM. The results of this model were. not good. Infact, it yielded a correlation of .045.
    - Part B
      - Part A was replicated using a sigmoid kernel. In this case the results after predicting with the model did not improve. The correlation did change but it changed to -0.006.
      - Part A was replicated using the polynomial kernel. The results after predicting demonstrated a slightly better result with the correlation increasing to .054.
  - **Q2: Neural Networks**
    - Three Neural Network models were created in order to determine the best one. The first neural network contained only 1 hidden node, while the second and third contained 5 and 10 respectively. The model that demonstrated the best results was the model with 10 hidden nodes. The correlation, activation function and number of neurons per layer for each model are as follows:
      - 1 hidden neuron
        - Correlation:
          -
```
> cor(predicted_favs,   fav_norm_test$favorite_count)
          [,1]
[1,] 0.1475258
```
        - Activation Function:
          -
```
> favorite_model$act.fct
function (x)
{
    1/(1 + exp(-x))
}
```
        - Number of Neurons per Layer:

        - Layer 1: 7
        - Layer 2: 2
- 5 hidden neurons:
    - Correlation:
        - 0.1104916
    - Activation Function:
        -
```
> favorite_model$act.fct
function (x)
{
    1/(1 + exp(-x))
}
```
    - Number of Neurons per Layer:
        - Layer 1: 35
        - Layer 2: 6
- 10 hidden neurons:
    - Correlation:
        - 0.1736068
    - Activation Function:
        -
```
> favorite_model$act.fct
function (x)
{
    1/(1 + exp(-x))
}
```
    - Number of Neurons per Layer:
        - Layer 1: 70
        - Layer 2: 11
    - Graph:



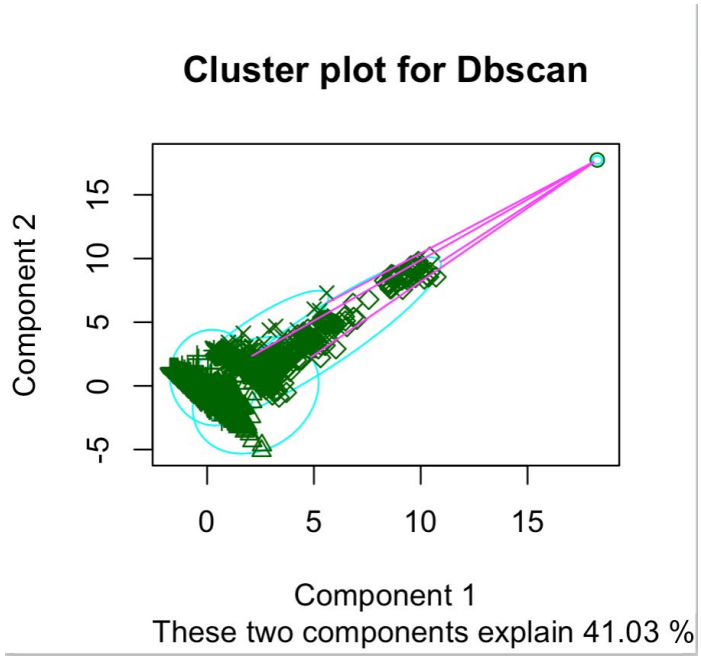
- **Q3: Clustering**
  - Predictor variable was divided into different classes similar to milestone 2 (0, 10, 100, 1000, 10000). The division was chosen for its logarithmic values. It was also noted after some analysis of the data that this perfectly encapsulated the outliers in the data without isolating the cases where likes or retweets were low. For example, 0-10 likes or retweets could be seen as normal, 10-100 as medium, 100-1000 as high and so forth.
  - K means, Spectral and Mclust clustering were used in order to create clustering models. The number of classes, number of elements per class and structural plots are as follows:
    - K means:
      - Number of classes:
        - 5 classes
      - Number of elements per class:
        - [3588, 4517, 2212, 1169, 3550]
      - Structural plots:

- 

- 

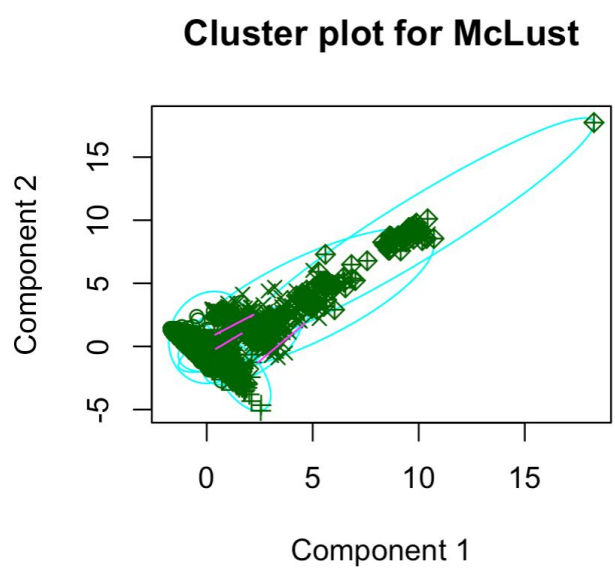
- Spectral:
  - Number of classes:
    - 4
  - Number of elements per class:
    - [8104, 5762, 180, 989]
  - Structural plots:

-



- Mclust:
  - Number of classes:
    - 9
  - Number of elements per class:
    - [4429, 1180, 932, 1019, 4443, 539, 794, 1549, 151]
  - Structural plots:

-

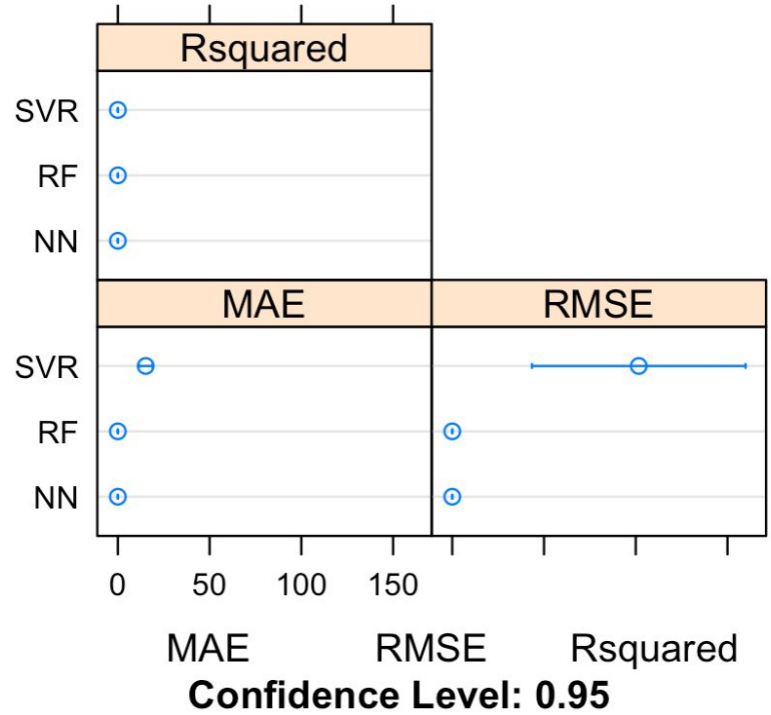


- **Q4: Comparative Analysis**
  - Comparative analysis was performed using repeated cross validations, cross validation and bootstrapping methods. For all three cases the results were similar and can be summarized by observing the results of the repeated cross validation below.

```
MAE
             Min.      1st Qu.      Median         Mean      3rd Qu.         Max. N
SVR 1.296013e+01 13.229273583 15.26120073 15.185782053 17.217709201 17.260593759
NN  9.234959e-04  0.001090695  0.00123441  0.001196151  0.001339866  0.001392288
RF  1.240954e-03  0.001242006  0.00124524  0.001252998  0.001256232  0.001280559

RMSE
             Min.      1st Qu.       Median         Mean      3rd Qu.         Max.
SVR 63.216911478 74.179103380 103.35245005 101.65424967 130.82759634 136.69518708
NN   0.007070234  0.007138265   0.01029423   0.01030192   0.01345789   0.01354899
RF   0.007500548  0.007796014   0.01044792   0.01043566   0.01308757   0.01334627

Rsquared
             Min.     1st Qu.      Median       Mean    3rd Qu.       Max. NA's
SVR 3.702154e-03 0.004118390 0.008727571 0.01073185 0.01534103 0.02177011    0
NN  7.024277e-06 0.002079271 0.005573401 0.00755044 0.01104457 0.01904793    0
RF  1.098781e-02 0.016134400 0.021268851 0.02083550 0.02596995 0.02981647    0
```



- **Interpretation:**
  - The interpretations for the comparisons are very simple. None of the models created could accurately predict any of the values that we were looking for. The actual prediction performance was very poor for all three models as demonstrated by the R squared value and the RMSE values for all three models. Though, SVR did demonstrate better performance than the others, it was still very poor. I do not believe that feature selection would play a role to improve performance seeing as how in previous multivariate regressions only few features were shown to be significant. In this case those have been removed as to not provide overlap (for the likes prediction the best predictor is amount of retweets, but we cannot keep the retweets as we are testing for that as well). This question just demonstrates that at least with the metadata of the tweet alone the amount of likes cannot be predicted.

- **Tweet Retweets**
  - **Q1: SVM**
    - Part A

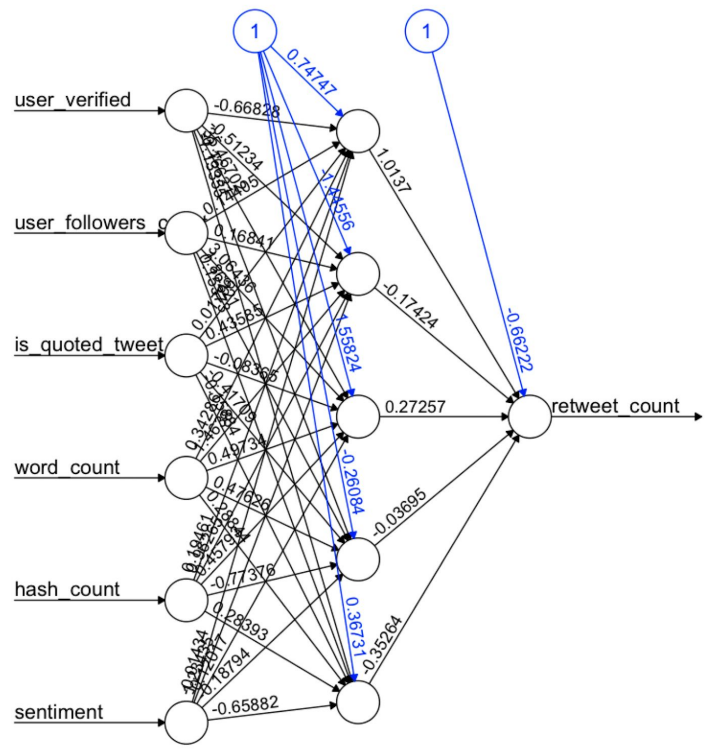
- Since this is a regression problem a SVR model had to be created instead of an SVM. The results of this model were not good. Infact, it yielded a correlation of 0.05312411.

- Part B
  - Part A was replicated using a sigmoid kernel. In this case the results after predicting with the model did not improve. The correlation did change but it changed to -0.006180538.
  - Part A was replicated using the polynomial kernel. The results after predicting demonstrated a slightly worse result with the correlation decresing to 0.03860751.

- **Q2: Neural Networks**
  - Three Neural Network models were created in order to determine the best one. The first neural network contained only 1 hidden node, while the second and third contained 5 and 10 respectively. The model that demonstrated the best results was the model with 10 hidden nodes. The correlation, activation function and number of neurons per layer for each model are as follows:
    - 1 hidden neuron
      - Correlation:
        -
```
> cor(predicted_favs,   fav_norm_test$retweet_count)
          [,1]
[1,] 0.1376953
```
      - Activation Function:
        -
```
> favorite_model$act.fct
function (x)
{
    1/(1 + exp(-x))
}
```
      - Number of Neurons per Layer:
        - Layer 1: 35
        - Layer 2: 6
    - 5 hidden neurons:
      - Correlation:
        -
```
> cor(predicted_favs,   fav_norm_test$retweet_count)
           [,1]
[1,] 0.07699982
```
      - Activation Function:

```
> favorite_model$act.fct
function (x)
{
    1/(1 + exp(-x))
}
```

- Number of Neurons per Layer:
  - Layer 1: 35
  - Layer 2: 6
- Graph:



- 10 hidden neurons:
  - Correlation:

```
> cor(predicted_favs,   fav_norm_test$retweet_count)
           [,1]
[1,] 0.07685325
```

  - Activation Function:

```
> favorite_model$act.fct
function (x)
{
    1/(1 + exp(-x))
}
```

- Number of Neurons per Layer:
  - Layer 1: 70
  - Layer 2: 11

- **Q3: Clustering**
  - Predictor variable was divided into different classes similar to milestone 2 (0, 10, 100, 1000, 10000). The division was chosen for its logarithmic values. It was also noted after some analysis of the data that this perfectly encapsulated the outliers in the data without isolating the cases where likes or retweets were low. For example, 0-10 likes or retweets could be seen as normal, 10-100 as medium, 100-1000 as high and so forth.
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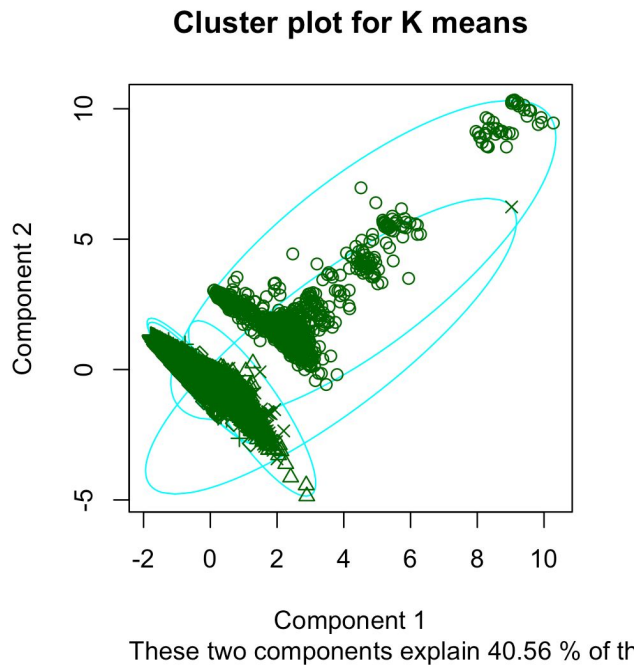
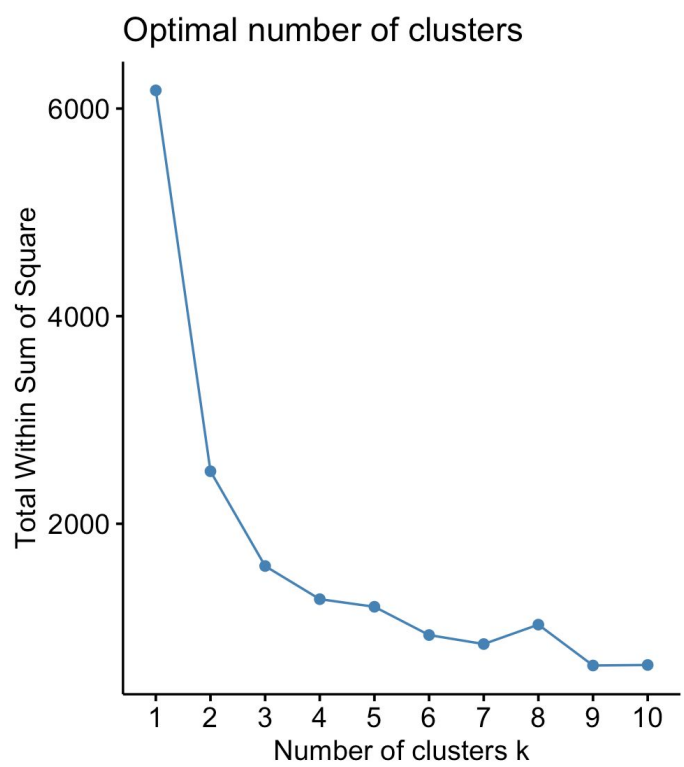
      - Structural plots:

- 

- 

- Spectral:
  - Number of classes:
    - 4
  - Number of elements per class:
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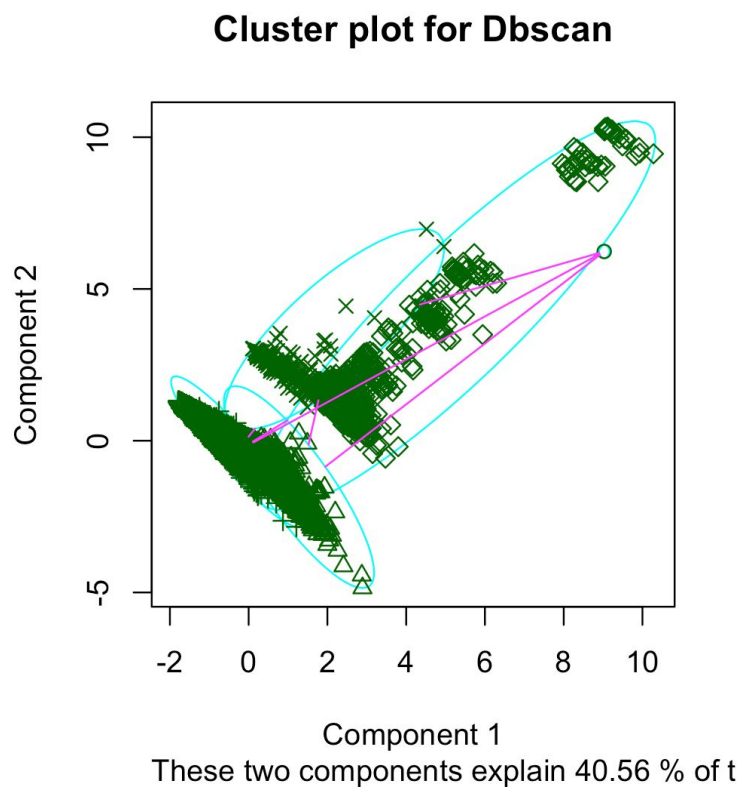
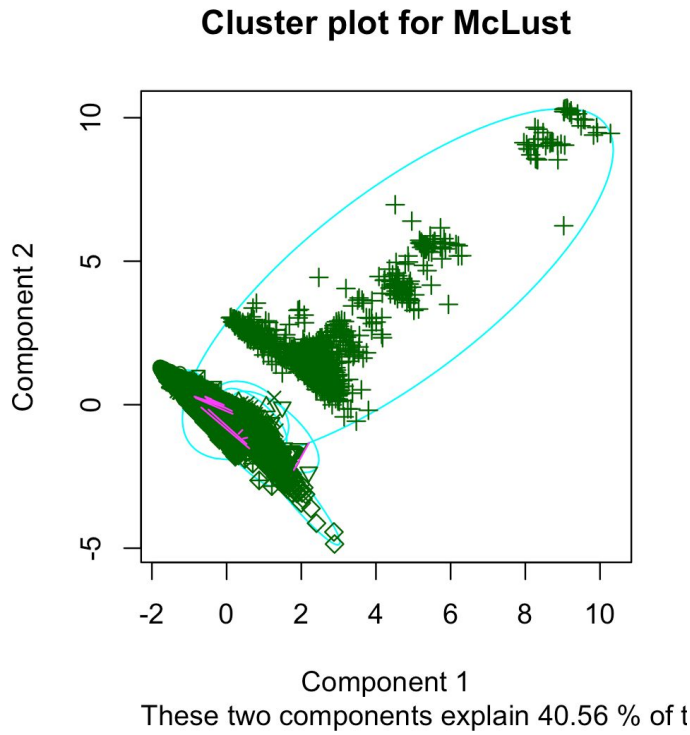
- Structural plots:
  -



- Mclust:
  - Number of classes:
    - 9
  - Number of elements per class:
    - [4006, 1056, 1170 ,4381, 953, 1714, 1330, 90, 336]
  - Structural plots:

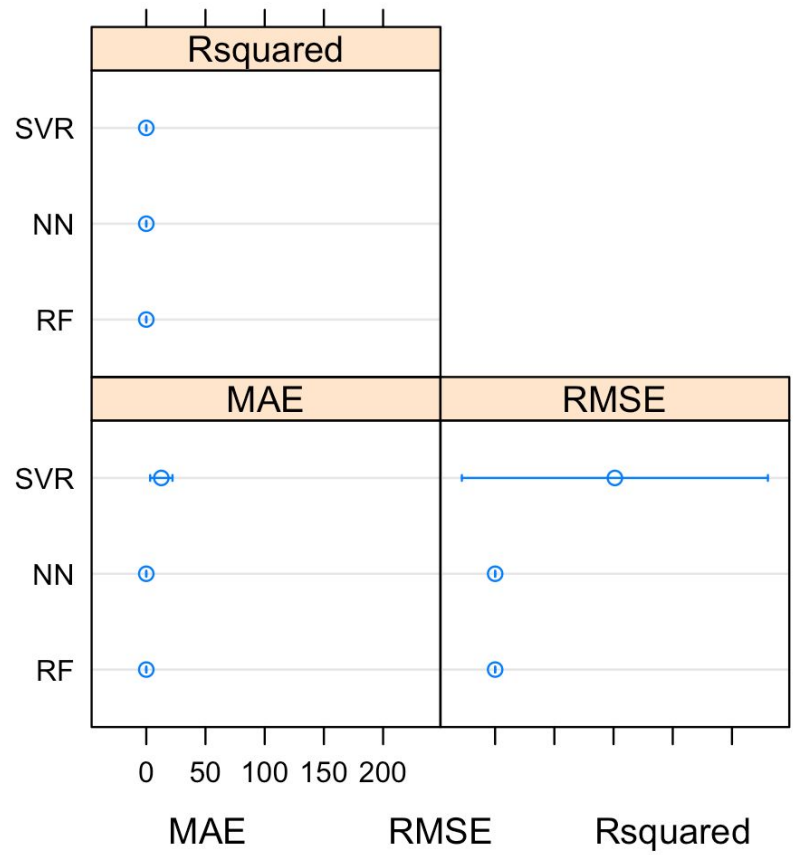
-



-

- **Q4: Comparative Analysis**
  - Comparative analysis was performed using repeated cross validations, cross validation and bootstrapping methods. For all three cases the results were similar and can be summarized by observing the results of the repeated cross validation below.

```
MAE
             Min.      1st Qu.       Median         Mean      3rd Qu.         Max.
SVR 7.5257200354 7.6018882709 1.271089e+01 1.270484e+01 1.781384e+01 1.787186e+01
NN  0.0006084779 0.0006150509 6.267895e-04 6.264474e-04 6.381860e-04 6.437327e-04
RF  0.0004655721 0.0004878390 5.003628e-04 4.978520e-04 5.103759e-04 5.251104e-04
    NA's
SVR    0
NN     0
RF     0

RMSE
             Min.      1st Qu.       Median         Mean      3rd Qu.         Max.
SVR 29.400094787 31.520567061 1.017046e+02 1.011318e+02 171.31585590 171.71788762
NN   0.002574792  0.002747146 7.218843e-03 7.174446e-03   0.01164614   0.01168531
RF   0.002696843  0.002778330 7.282306e-03 7.258097e-03   0.01176207   0.01177093
    NA's
SVR    0
NN     0
RF     0

Rsquared
             Min.      1st Qu.      Median        Mean     3rd Qu.        Max. NA's
SVR 0.0002276603 0.0002891738 0.006162220 0.007646387 0.013519433 0.018033449    0
NN  0.0004001473 0.0005245086 0.003646129 0.004083116 0.007204737 0.008640059    0
RF  0.0008572203 0.0010586806 0.003110635 0.003812151 0.005864106 0.008170112    0
```



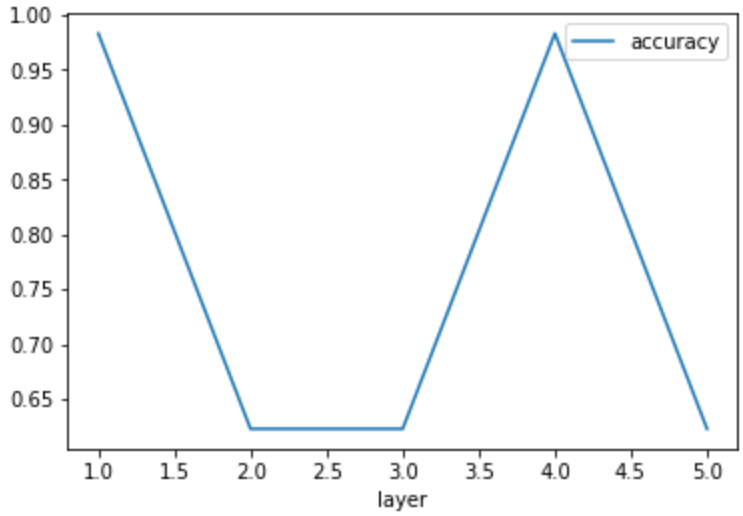
- Interpretation:
  - The interpretations for the comparisons are very simple. None of the models created could accurately predict any of the values that we were looking for. The actual prediction performance was very poor for all three models as demonstrated by the R squared value and the RMSE values for all three models. I do not believe that feature selection would play a role to improve performance seeing as how in previous multivariate regressions only few features were shown to be significant. In this case those have been removed as to not provide overlap (for the likes prediction the best predictor is amount of retweets, but we cannot keep the retweets as we are testing for that as well). This question just demonstrates that at least with the metadata of the tweet alone the amount of retweets cannot be predicted.

RESEARCH QUESTION 2 - Classifying Conspiracy Tweets

- **Q1: SVM**
  - My dataset contains only two categories: Conspiracy and Non-conspiracy. As the labels are discrete, I couldn't turn them into multi-class problems. So, here I am reporting the analysis for binary classification only.
  - I have used 3 different kernels for SVM. They are: Radial Basis Function (RBF), Linear and Polynomial. RBF and polynomial kernels show the same performance and they perform better than linear.

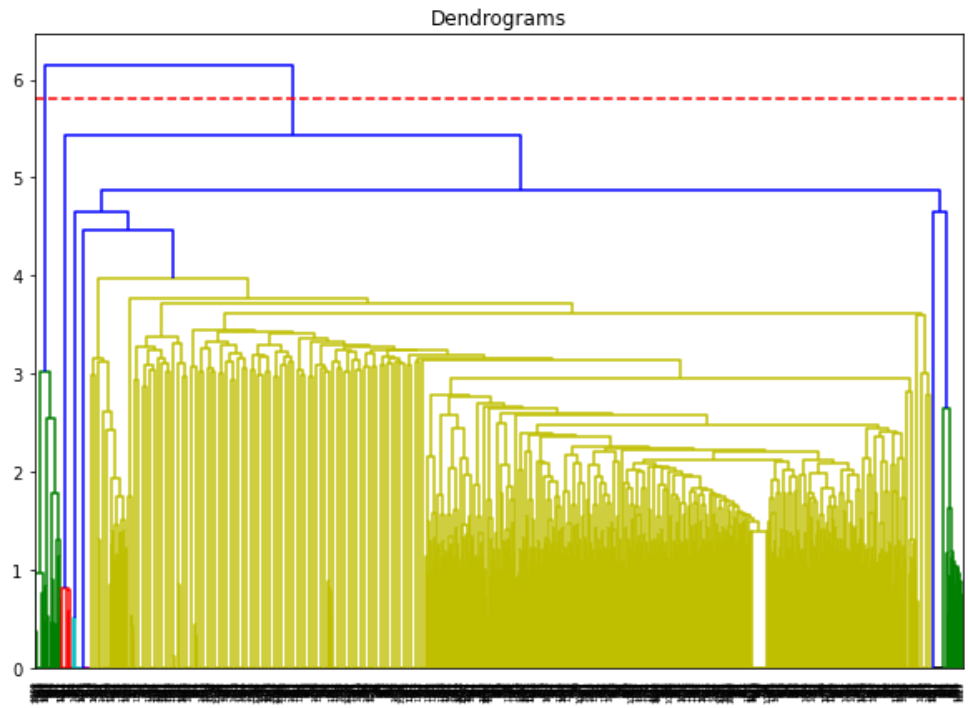
Kernel	Precision	Recall	F1	Accuracy	Specificity
Radial Basis Function (RBF)	0.99	0.99	0.99	0.99	0.98
Linear	0.95	0.95	0.95	0.95	0.96
Polynomial	0.99	0.99	0.99	0.99	0.98

- **Q2: Neural Networks**
  - I have used the Recurrent Neural Network. To implement this, I have used a python package named keras. After tokenizing and padding the input sequence, it has been fed to the model. For calculating loss, I have used categorical cross entropy and for validation, I have used accuracy metrics. The maximum number of the features has been set to 3000 and input and out nodes were set to 200 respectively. I have used 'softmax' for the activation. For different numbers of hidden layers, I have evaluated the performance. The following figure shows the performance comparison. For 1 and 4 hidden layers, the neural network performed best with accuracy 98.32%.



- **Q3: Clustering**
  - I used three unsupervised clustering methods: kMeans, Hierarchical Clustering, and DBScan.
  - For kMeans, I set the number of clusters to 2, as I have two categories. The initial center was chosen randomly. First cluster contains 890 elements, and the second one contains 297 elements. As the dataset I used has multi-dimensions, I didn't plot them to visualize the cluster. The top words from cluster one are: *coronavirus, china, virus, num, amp, coronavirusoutbreak, wuhan, news, people, trump*. From the second cluster, the top words are: *coronavirus, chinese, doctor, china, wuhan, truth, li, numnum, died*. The second cluster includes some words like doctor, died, china, etc. which may indicate a popular conspiracy topic related to the dying of a doctor in China who identified the coronavirus first.
  - For hierarchical clustering, I have also chosen two clusters. First cluster contains 1153 elements and the second one contains 34. The following

figure shows the hierarchical diagram.



- For DBScan, I set the minimum number of samples to 6. Maximum number of elements are considered as noise. Apart from that, I got ten clusters having a few elements for each cluster.
  C_0: 13, C_1: 10, C_2: 13, C_3: 20, C_4: 6, C_5: 11, C_6: 16, C_7: 6, C_8: 6, C_9: 8

- **Q4: Comparative Analysis and Feature Selection**
  - I have used three cross-validation techniques: kFold, Leave-p-out, and Random Subsampling. For each technique, the number of splits has been set to 5.
  - For different validation techniques, I compared the performance of the models (SVM, Neural network from Milestone 3 and Random Forest, Decision Tree from Milestone 2). The following table shows the comparison of their performances. In kFold validation, the SVM outperforms the models. The neural net is lagging behind the classical ML models because of the low number of training samples.

CV	model	precision	recall	f1	accuracy
kFold	SVM	**0.993**	**0.993**	**0.993**	**0.993**
	Decision Tree	0.966	0.979	0.971	0.973
	Random Forest	0.99	0.994	0.992	0.992
	Neural Net	0.955	0.972	0.962	0.965
LeavePOut	SVM	0.988	0.992	0.99	0.991
	Decision Tree	0.981	0.987	0.984	0.985
	Random Forest	0.988	0.992	0.99	0.99
	Neural Net	0.972	0.981	0.976	0.977
Random SubSampling	SVM	0.981	0.971	0.976	0.977
	Decision Tree	0.956	0.963	0.959	0.961
	Random Forest	0.98	0.975	0.977	0.979
	Neural Net	0.951	0.96	0.955	0.957

- I have also checked the performance of the models after applying feature selection techniques. For selecting the features, I have used SVM and then different models have been applied on the selected features for classification. The following table shows the performance comparison for kFold validation techniques. It seems like after applying feature selection, all models performed less than the original state.

	precision	recall	f1	accuracy	model	CV
0	0.954980	0.970127	0.961260	0.963773	SVM	kFold
1	0.964621	0.976893	0.969702	0.971347	Decision Tree	kFold
2	0.984685	0.990531	0.987348	0.988196	Random Forest	kFold

RESEARCH QUESTION 3- Classifying Fear Causing/Relieving Tweets

- **Q1: SVM:** Since this is a classification problem, the result categories were only two.. For part A, there is only binary classification.
  Part B: The classifications used was C-SVC. The kernels used were linear and Gaussian Radial.

  - **AGREEMENT_RBF**

```
FALSE      TRUE
127        124
0.5059761   0.4940239
```

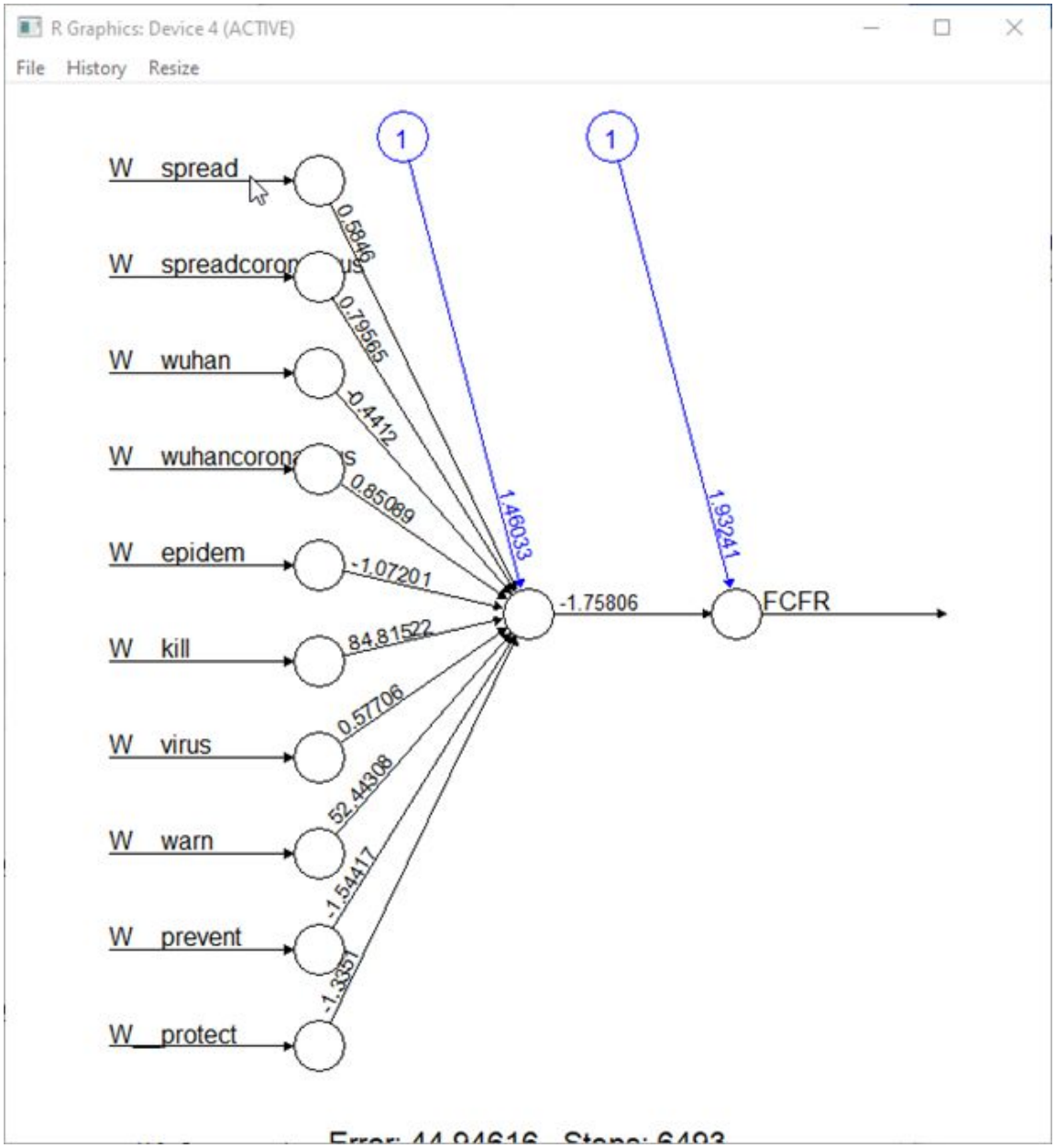
  - **FEAR_CLASSIFIER**

```
Support Vector Machine object of class "ksvm"

SV type: C-svc  (classification)
 parameter : cost C = 1

Linear (vanilla) kernel function.

Number of Support Vectors : 210

Objective Function Value : -131.4761
Training error : 0.112
```

  - **FEAR_CLASSIFIER_RBF**

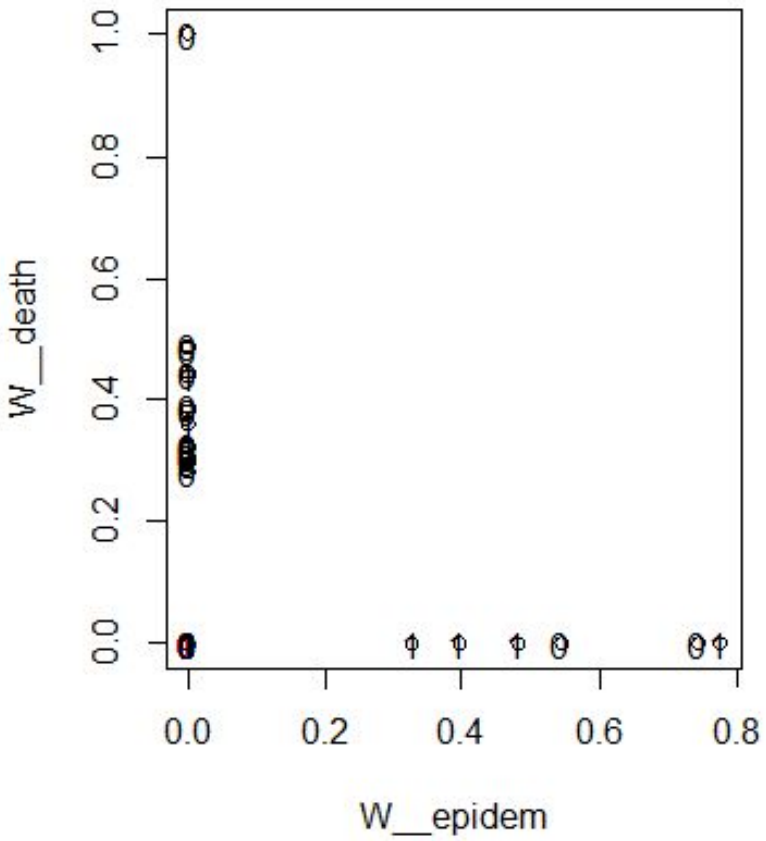
```
Support Vector Machine object of class "ksvm"

SV type: C-svc (classification)
 parameter: cost C = 1

Gaussian Radial Basis kernel function.
 Hyperparameter : sigma =  0.00150949417176231

Number of Support Vectors : 237

Objective Function Value : -220.6617
Training error: 0.436
```

- **Q2: Neural Networks**

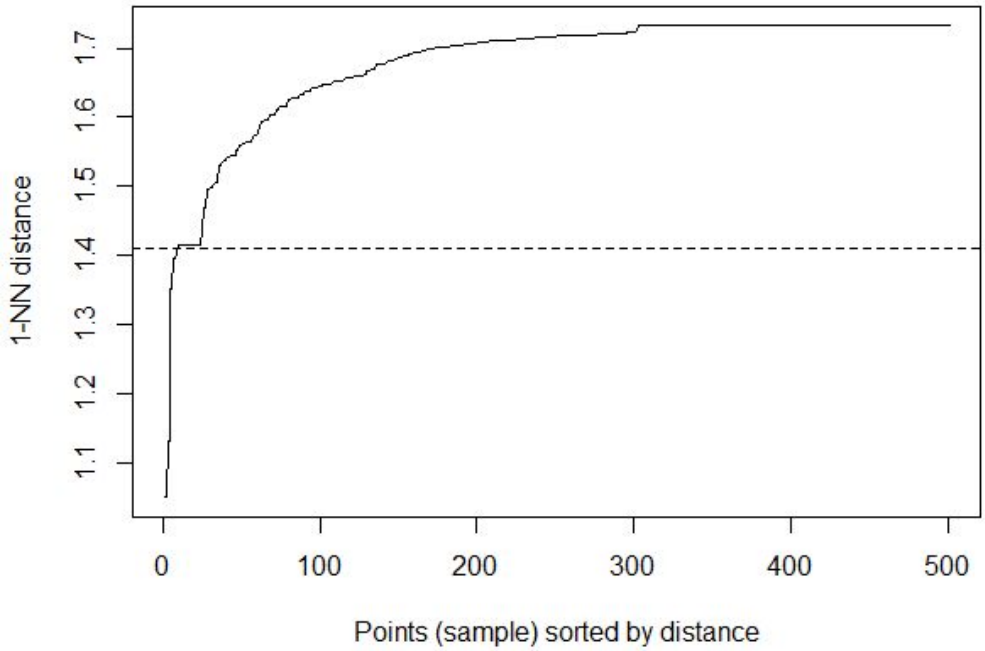


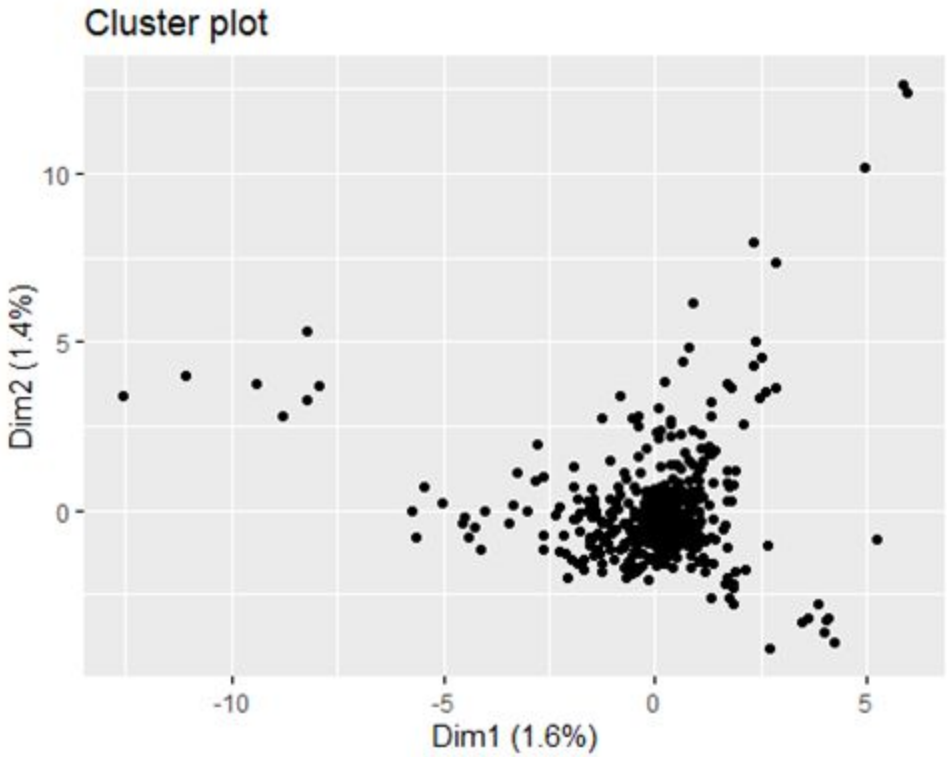
- **predicted_FCFR <-model_results$net.result: [,1]**
- **cor(predicted_FCFR,fear_test$FCFR): [1,] 0.9334275**

- **Q3: Clustering**
  - **K-Means**

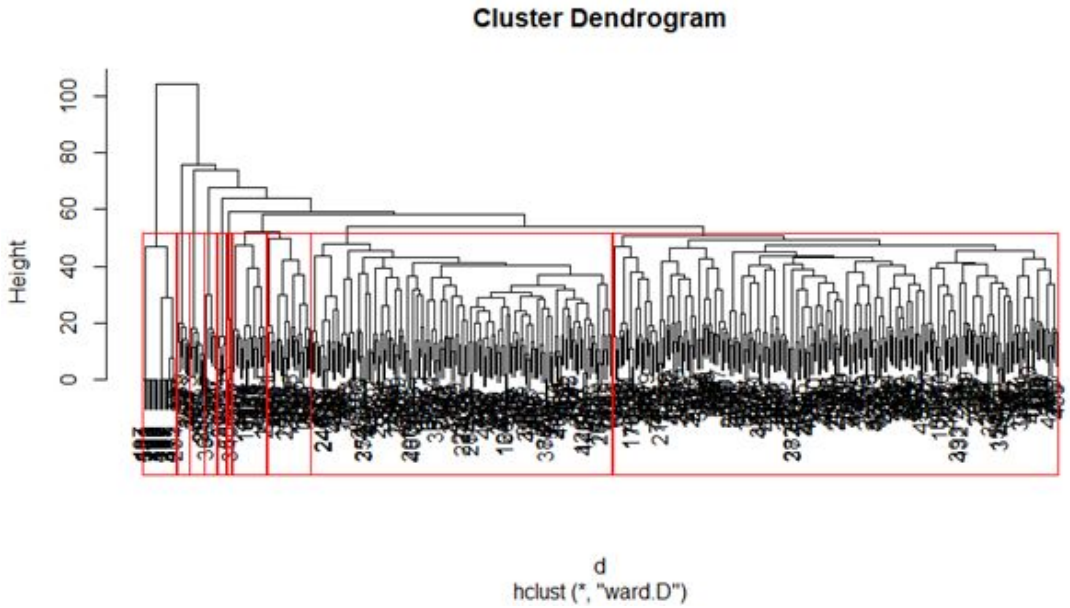

W__death
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.2
0.4
0.6
0.8
W__epidem

- **Density Based**





- **Hierarchical (Dendogram)**



- **Q4: Comparative Analysis**

Support Vector Machines with Linear Kernel

351 samples
196 predictors
2 classes: 'C', 'R'

Pre-processing: centered (196), scaled (196)
Resampling: Cross-Validated (10 fold, repeated 3 times)
Summary of sample sizes: 316, 317, 316, 317, 315, 315, ...
Resampling results:

Accuracy	Kappa
0.5456365	0.09063851

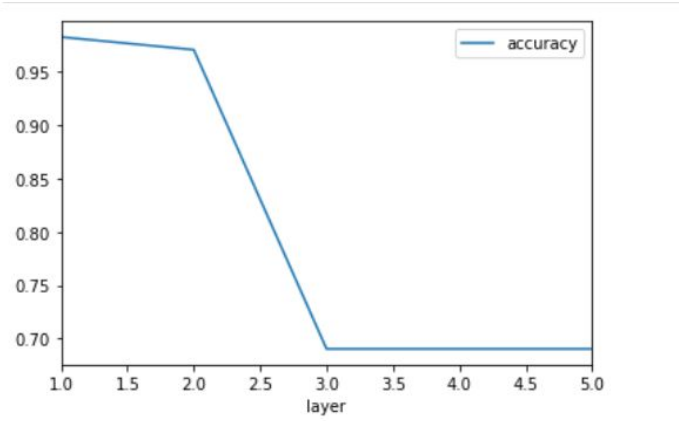
Tuning parameter 'C' was held constant at a value of 1

RESEARCH QUESTION 4 - Classifying Information Tweets

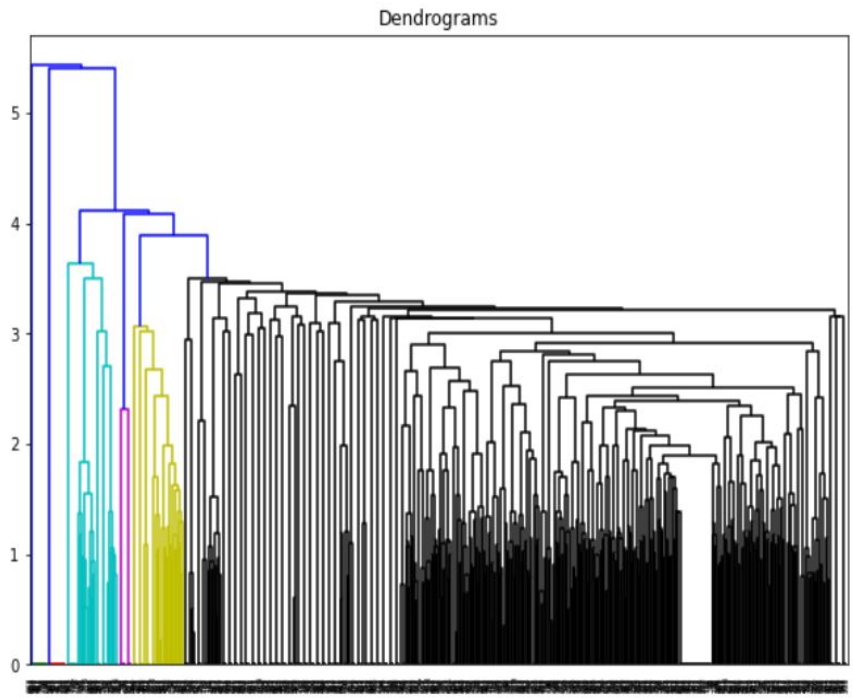
- **Q1: SVM**
  - **PART A:** Since this is a classification problem, the result categories were only two. For part A, there is only binary classification.
  - **PART B:** In this part, 3 different kernels were used for SVM. They are: Radial Basis Function (RBF), Linear and Polynomial. RBF and polynomial kernels show the same performance and they perform better than linear.
  - From the table, we can see that while the Linear kernel performs quite well, the other two seem to be overtrained. This could be due to the small size of the dataset and repetitions.

Kernel	Precision	Recall	F1	Accuracy	Specificity
Radial Basis Function (RBF)	1.0	1.0	1.0	1.0	1.0
Linear	0.98	0.99	0.99	0.99	0.98
Polynomial	1.0	1.0	1.0	1.0	1.0

- **Q2: Neural Networks**
  - RNN or Recurrent Neural Networks from the Python library keras was used.
  - For Neural Networks, the input and output nodes were kept at 200 and the softmax activation was used.
  - For calculating loss, categorical cross entropy was used and for validation, accuracy metrics.
  - We plot the accuracy for each hidden layer and we can see from the following graph, that the first two layers perform better than the rest. The best performance was given by the first layer: an accuracy of 98.2%.



- **Q3: Clustering**
  - Three clustering algorithms were used: k-Means, Hierarchical Clustering, DBScan.
  - or kMeans, the number of clusters set was 2, as we classify our data into 2 categories. The initial center was chosen randomly. First cluster contains 505 elements, and the second one contains 165 elements. As the dataset I used has multi-dimensions, we didn't plot them to visualize the cluster. The top words from cluster one are: *coronavirus, numnum, truth, num, people, new, just, hospital, australia.* From the second cluster, the top words are: *china, virus, coronavirus, wuhan, outbreak, corona, details, gt, health, novel.*
  - For hierarchical clustering, again the number of clusters was kept the same. We got 655 elements in the first cluster and 15 in the second. The following plot depicts the hierarchy



- For DBScan, the minimum number of samples was set to 6. Maximum number of elements are considered as noise. Apart from that, we got 5 clusters having a few elements for each cluster.
  C_0: 9, C_1: 27, C_2: 15, C_3: 6, C_4: 15

- **Q4: Comparative Analysis and Feature Selection**
  - Three cross-validation techniques were used: kFold, Leave-p-out, and Random Subsampling. For each technique, the number of splits has been set to 5.
  - For different validation techniques, I compared the performance of the models (SVM, Neural network from Milestone 3 and Random Forest, Decision Tree from Milestone 2). The following table shows the comparison of their performances. In kFold validation, the SVM outperforms the models. The neural net is lagging behind the classical ML models because of the low number of training samples.

CV	**model**	**precision**	**recall**	**f1**	**accuracy**
kFold	SVM	0.997	0.998	0.998	0.998
	Decision Tree	0.961	0.980	0.969	0.973

	Random Forest	0.995	0.997	0.996	0.997
	Neural Net	0.982	0.991	0.986	0.988
LeavePOut	SVM	0.977	0.944	0.958	0.966
	Decision Tree	0.903	0.912	0.907	0.922
	Random Forest	0.974	0.942	0.955	0.964
	Neural Net	0.968	0.967	0.967	0.972
Random SubSampling	SVM	0.980	0.956	0.966	0.971
	Decision Tree	0.944	0.952	0.948	0.954
	Random Forest	0.981	0.959	0.968	0.973
	Neural Net	0.931	0.953	0.940	0.946

- I have also checked the performance of the models after applying feature selection techniques. For selecting the features, I have used SVM and then different models have been applied on the selected features for classification. The following table shows the performance comparison for kFold validation techniques. It seems like after applying feature selection, all models performed less than the original state.

	precision	recall	f1	accuracy	model	CV
0	0.970935	0.982713	0.976123	0.979104	SVM	kFold
1	0.972103	0.985755	0.978080	0.980597	Decision Tree	kFold
2	0.982913	0.992610	0.987487	0.989552	Random Forest	kFold